

DEEPANSHU MANGOLIA

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EDUCATION

Bachelor of Technology (Computer Engineering), Dayalbagh Educational Institute 2021–2024
CGPA: 7.90

Diploma in Engineering (Computer Science), Dayalbagh Educational Institute 2018–2021
CGPA: 8.61

SKILLS

Programming Languages	C++, Python, SQL
Quant/Market Data	TimescaleDB, PostgreSQL, Upstox API, NSE Data APIs
Data & Automation	Pandas, NumPy, Requests, BeautifulSoup, IMAP, Cronjobs
ML & Analytics	Scikit-learn, TensorFlow, OpenCV
Systems & Tools	Linux, Git, Tmux, Microsoft Office

EXPERIENCE

Quant Developer Jul 2024 – Present
Raghunandan Money, Uttar Pradesh, India

- **Market Data Engineering** — Built automated pipelines for NSE Bhavcopy (Cash, Indices, F&O), including a published PyPI library [Rmoney.bhavcopy](#) used for production-grade Bhavcopy ingestion. Automated 10+ years of historical data, corporate action adjustments, Upstox market data, and index constituent ingestion, achieving **100% automation** and delivering high-quality datasets for quant research and trading models.
- **Lean (QuantConnect) Framework Integration** — Adapted and optimized the Lean backtesting framework for Indian markets by designing high-speed data ingestion pipelines from internal datasets into Lean-compatible formats. Implemented holiday calendars, partial session logic, symbol mapping, and corporate action adjustments, resulting in significantly improved backtesting accuracy and dramatically reduced load times.
- **High-Performance Time-Series Systems** — Engineered optimized time-series data-access libraries using hypertable partitioning and SQL tuning, improving query performance by **80%**. Developed a multithreaded C++/Python TBT → OHLC candle engine, reducing candle generation time by **70%** for high-resolution datasets.
- **Regulatory Intelligence Automation** — Developed a **Regulatory Intelligence Automation** system with automated NSE circular retrieval, PDF extraction, summarization, and scheduled email distribution, streamlining compliance monitoring and reducing manual oversight effort by **80%**.
- **Automated Trading Infrastructure** — Built and deployed an automated execution system using XTS API for order placement, routing, and management, increasing execution reliability and supporting live trading strategies.

Tech Stack: Python, C++, Pandas, NumPy, PostgreSQL, TimescaleDB, Linux, Cronjobs, Requests, BeautifulSoup, PyPDF, IMAP, SMTP, OpenAI API, Upstox API, XTS API

Java Developer Intern Apr 2023 – Aug 2023
Genisup India Pvt. Ltd., Uttar Pradesh, India

- Selected for a **co-op program** under guidance from Microsoft experts; worked on advanced CS/NLP problem statements.
- Developed an **NLP-based text similarity engine** implementing Cosine Similarity and distance metrics from scratch using core OOP principles.
- Wrote **75+ optimized SQL queries** for backend validation, processing, and reporting.
- Built an Android application enabling file-to-file similarity comparison using the custom algorithm.
- GitHub: [Plagiarism Checker](#)

PROJECTS

SHORT STRANGLE — Developed and backtested a short-strangle options strategy; implemented historical options data fetch/filtering and execution logic with stop-loss. Analyzed **CAGR**, **max drawdown**, and **win rate**; produced equity curve and drawdown visualizations. *Tech*: Python, Pandas, NumPy, Matplotlib.

Trade Data Reconciliation System — End to end reconciliation between broker trade confirmations and client orders: parsed EML/Excel attachments, handled multi-broker partial matches, generated structured summaries, and exposed a **Flask API** with scheduled daily automation. *Tech*: Python, SQLite, Pandas, Flask, Excel, SMTP.

RESEARCH & PUBLICATIONS

Convolutional Neural Network-Based Automated Acute Lymphoblastic Leukaemia Detection and Stage Classification

Published in: **International Journal of Scientific Research in Computer Science and Engineering (IJS-RCSE)**

[View Publication](#)

- Achieved test accuracy **92.12%** (loss **0.2208**); built a web app for real-time visualization and classification.
- Tools: Python, TensorFlow, NumPy, OpenCV, Deep Learning.